

The Influence of User Demographics and Ad Design on Facebook Purchase Behavior

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Introduction

Since the rise of social media, advertising has shifted from traditional outlets like newspapers and billboards to digital platforms. Facebook, with its large user base and advanced advertisement tools, remains a dominant platform. This project analyzes how effective Facebook ad campaigns are and whether increased spending leads to more sales, especially when considering ad design, demographics, and user interests.

Using real-world campaign data from an anonymous organization, I will explore how variables like spending, age, gender, and user clicks contribute to ad success. I chose this topic because Facebook has been a dominant social media platform and played a key role in advancing algorithm-based advertising. I was particularly interested in how effective Facebook's algorithm is at helping companies target their desired audiences, and whether that targeting is reflected in more purchases.

Literature Review

Social media marketing has grown significantly over the past decade, with platforms like Facebook offering companies advanced marketing tools with detailed performance reports. These tools allow marketers to reach specific audiences and come up with strategies using real-time engagement data. Effective Facebook ads, and digital ads in general, often rely on three main factors: personalization, demographic targeting, and an attractive, compelling design.

Several studies have explored these factors from different perspectives. For example, Al Kurdi and Alshurideh (2021) examined how Facebook ads influenced women's purchasing decisions for cosmetic products. This research study has identified several influencing factors, which include credibility, informativeness, and entertainment value of ads. The study found that when users viewed ads to be trustworthy from a known brand, and informative, they were more likely to click and view the product and consider purchasing. The emotional appeal also improved how users viewed the brand and helped distinguish it from competitors.

Likewise, Serra Inci Celebi (2015) explored how social motivations impacted how university students interacted with Facebook ads. The study found that users who used the platform to stay connected with others or follow trends were more open to ads that reflected their personal interests. Ads that aligned with a user's lifestyle, values, or aesthetic preferences stood out the most. The study also mentioned that "interpersonal utility was considered as a social need that drives a person to belong to an online social group and to share information with them," which supports the idea that personalized content is more likely to succeed. A similar pattern was seen in Hamouda's (2018) research on Facebook ads in the tourism industry. That study applied Ducoffe's (1995) advertising value model and found that informativeness, entertainment, and credibility influenced how people judged ad quality. However, ad irritation, caused by repetition, poor formatting, or aggressive messaging, could turn users away. Hamouda also pointed out that users have a desire to belong to a community and to express themselves publicly and share experiences, have a more positive attitude towards the digital advertisements. The importance of social media ads in this sector was also emphasized, as "companies in travel industry spent \$5.96 billion on social media advertising in 2016" (H. Manel, 2018).

Collectively, these studies confirm that ad design, emotional tone, user motivations, and perceived value significantly impact the success of Facebook ads. Building on these discoveries, I aim to discover how user engagement and purchase intent are influenced by the emotional and visual design of social media ads. Based on this, my first research question asks: does personalizing Facebook ads to match users' interests and self-identity increase the likelihood of purchasing from the ad? I hypothesize that ads with personalized content will increase user engagement and purchases because they will appear to be more aligned with the users' interests, which will catch their attention and be more engaged. In addition, I also want to explore how different user characteristics influence ad effectiveness. This leads to my second research question: do factors like age, gender, or interests affect how well ads perform? I hypothesize that demographic targeting, such as age and gender, increases the effectiveness of Facebook ads and increases sales overall.

Methods

This project uses a dataset from Kaggle, published in 2020 by an anonymous company. It includes ad ID, age, gender, impressions, clicks, spend, total conversions, and approved purchases. Age groups range from 30 to 49 and are split by gender.

To answer the research questions, I focused on key variables such as age, gender, and clicks. The sample was already collected by the organization through Facebook's advertising system and reflects real user interaction with their ads. Using Tableau, I grouped the data by age and gender and created a clustered bar chart showing approved purchases by the demographic. I also created a pivot table to summarize total purchases and clicks by group and gender. These visuals helped identify which demographics engaged and purchased most frequently.

Results/Discussion

As seen in the PivotTable, the 30 - 34 age group showed the most confirmed purchases, totaling 494. Men accounted for 299 of these, and women contributed 195. This group also had many clicks, 9,483, which indicates strong engagement with the ad design. Users in this age range may be more responsive to ads due to a mix of social media familiarity and stable financial standing, considering the age range.

The 45–49 group followed with 208 purchases, led by females, who also had the highest number of clicks among all female groups, 9,441. However, the 35–39 group had a higher conversion rate, 207 purchases from 7,094 clicks, suggesting more deliberate purchasing. As seen in Figure 1, men aged 30-34 had the most confirmed purchases of 299. Across all groups, females clicked more frequently, but males generally made more purchases. This implies that while women are more interactive, men are slightly more trigger-happy or make quicker decisions on a purchase.

These findings suggest Facebook ads are most effective for users aged 30–39. Campaigns that targeted a specific gender could be useful, but more information in the dataset would be needed. Personalization likely increases engagement and purchases by making ads more relevant to the user.

Limitations in the dataset include the narrow age range, 30 to 49, which limits insight into younger or older users. The “interest” variable was also numeric, which lacked detail on users’ preferences or interests, like fitness or fashion. This limited deeper analysis of interest-based personalization. Also, the dataset lacked qualitative data like user attitudes, which could have added context about ad reception. Furthermore, integrating survey data or including context on the ad campaign design in future research could strengthen the understanding of user motivations for ad engagement on Facebook.

Conclusion

This study looked at how effective ad campaigns are on Facebook using real-world data. The analysis found that age and gender play a significant role in purchase outcomes. Users aged 30–39 demonstrated the highest engagement and conversion rates, with men making slightly more confirmed purchases despite women clicking more frequently. These patterns support the idea that demographic targeting enhances ad performance. Additionally, the results align with prior research showing that personalization and emotional design improve user interaction and engagement with digital ads. Overall, the findings highlight the continued value of Facebook’s advertising platform in driving user engagement and sales when ads are tailored to user identity.

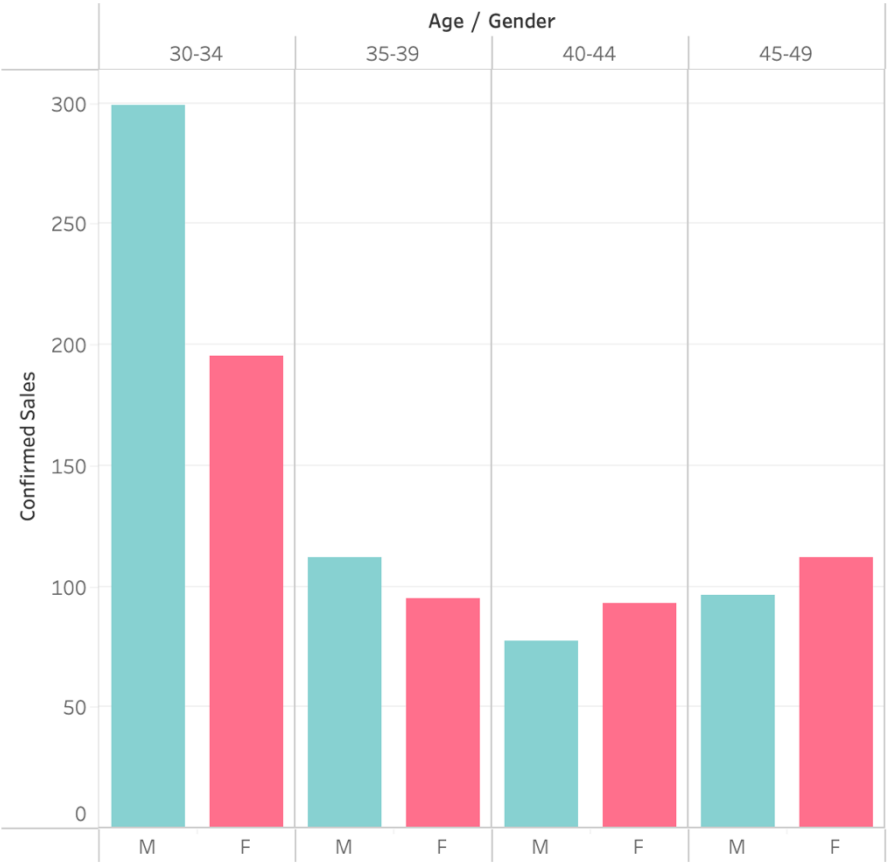
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Tables and Figures Here

Figure 1 – User Demographics and Ad Effectiveness

User Demographics and Ad Effectiveness



F		M		Total Clicks	Total Confirmed Sales	
Clicks	Confirmed Sales	Clicks	Confirmed Sales			
30-34	5099	195	4384	299	9483	494
35-39	4161	95	2933	112	7094	207
40-44	5177	93	2559	77	7736	170
45-49	9441	112	4411	96	13852	208